SOLUTIONS: SYSTEMS IDENTIFICATION

1. Solution

a) Based on the block diagram, one gets:

$$\begin{aligned} x_1(t+1) &= 0.5x_1(t) + a_1x_2(t) + x_3(t), \\ x_2(t+1) &= -0.5x_2(t) + 0.1x_4(t) + v_1(t), \\ x_3(t+1) &= a_2x_3(t) + x_4(t) + v_2(t), \\ x_4(t+1) &= a_3x_2(t) + 0.5x_4(t). \end{aligned}$$

[3 Marks]

b) Referring to the answer to Question 1-a), it follows that the second and fourth equations evolve independently from the first and the third ones. Hence, we consider these equations separately:

$$\begin{aligned} x_2(t+1) &= -0.5x_2(t) + 0.1x_4(t) + v_1(t), \\ x_4(t+1) &= a_3x_2(t) + 0.5x_4(t). \end{aligned} \tag{1.1}$$

According to (1.1), the stochastic processes $x_2(\cdot)$ and $x_4(\cdot)$ are stationary if all eigenvalues of the matrix

$$M = \begin{bmatrix} -0.5 & 0.1 \\ a_3 & 0.5 \end{bmatrix}$$

lie strictly inside the unit circle. To analyze the eigenvalues of M, we can investigate the roots of $\det(zI - M) = 0$:

$$(z+0.5)(z-0.5) - a_3/10 = 0,$$

or

$$z^2 - 1/4 - a_3/10 = 0.$$

Therefore, the following values for a_3 are acceptable:

$$0 \leq \frac{1}{4} + \frac{a_3}{10} < 1$$

which yields

$$\frac{-10}{4} \leq a_3 < \frac{30}{4}.$$

Now, since $x_4(\cdot)$ is a stationary process, also $x_4(\cdot) + v_2(\cdot)$ is stationary. Then, we let $\hat{v}_2(\cdot) = x_4(\cdot) + v_2(\cdot)$, and we write

$$x_3(t+1) = a_2x_3(t) + \underbrace{x_4(t) + v_2(t)}_{\hat{v}_2(t)}.$$

Therefore, it can be said that $x_3(\cdot)$ is also a stationary process if

$$a_2 < 1.$$

Answers

Now, since $x_2(\cdot)$ and $x_3(\cdot)$ are stationary process, let us consider the first equation as

$$x_1(t+1) = 0.5x_1(t) + \underbrace{a_1x_2(t) + x_3(t)}_{n(t)}$$

where $n(\cdot)$ is a stationary process. Therefore, as $|0.5| < 1$, for any finite a_1, one can conclude that $x_1(\cdot)$ is a stationary process as well.

[6 Marks]

c) According to the answer to Question 1-b), for $a_1 = a_2 = a_3 = 0$, $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$ are stationary processes. Therefore, $\mathbb{E}[x_i(t+1)] = \mathbb{E}[x_i(t)], i = 1,2,3,4$. Thus, one obtains:

$$\begin{aligned}
m_1 &= 0.5m_1 + m_3,\\
m_2 &= -0.5m_2 + 0.1m_4 + \mathbb{E}[v_1(t)],\\
m_3 &= m_4 + \mathbb{E}[v_2(t)],\\
m_4 &= 0.5m_4
\end{aligned}$$

which can be simplified as

$$\begin{aligned}
m_1 &= 0.5m_1 + m_3,\\
m_2 &= -0.5m_2 + 0.1m_4,\\
m_3 &= m_4 + 1,\\
m_4 &= 0.5m_4.
\end{aligned}$$

Therefore, we get $m_1 = 2$, $m_2 = 0$, $m_3 = 1$, and $m_4 = 0$.

[3 Marks]

d) Let us define

$$\begin{aligned}
\tilde{x}_1(t) &= x_1(t) - m_1,\\
\tilde{x}_2(t) &= x_2(t) - m_2,\\
\tilde{x}_3(t) &= x_3(t) - m_3,\\
\tilde{x}_4(t) &= x_4(t) - m_4,\\
\tilde{v}_1(t) &= v_1(t),\\
\tilde{v}_2(t) &= v_2(t) - 1.
\end{aligned}$$

Therefore, from the answer to Question 1-a), for $a_1 = a_2 = a_3 = 0$, we obtain

$$\begin{aligned}
\tilde{x}_1(t+1) &= 0.5\tilde{x}_1(t) + \tilde{x}_3(t),\\
\tilde{x}_2(t+1) &= -0.5\tilde{x}_2(t) + 0.1\tilde{x}_4(t) + \tilde{v}_1(t),\\
\tilde{x}_3(t+1) &= \tilde{x}_4(t) + \tilde{v}_2(t),\\
\tilde{x}_4(t+1) &= 0.5\tilde{x}_4(t).
\end{aligned} \tag{1.2}$$

Now, as the processes $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$ are stationary, we write

$$\mathbb{E}[\tilde{x}_4(t+1)^2] = \mathbb{E}[\tilde{x}_4(t)^2]$$

Answers

which from (1.2), gives

$$\mathbb{E}[\tilde{x}_4(t)^2] = 0.25\mathbb{E}[\tilde{x}_4(t)^2],$$

implying that

$$\mathbb{E}[\tilde{x}_4(t)^2] = 0. \tag{1.3}$$

In a similar way, one obtains

$$\mathbb{E}[\tilde{x}_2(t+1)^2] = \mathbb{E}[\tilde{x}_2(t)^2]$$

which from (1.2), gives (note that there is no correlation between $\tilde{x}_i(t)$ and $\tilde{v}_i(t)$)

$$\mathbb{E}[\tilde{x}_2(t)^2] = 0.25\mathbb{E}[\tilde{x}_2(t)^2] + 0.01\mathbb{E}[\tilde{x}_4(t)^2] + \mathbb{E}[v_1(t)^2] - 0.1\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)].$$

In turn, from (1.3) and since $\mathbb{E}[v_1(t)^2] = 1$, it follows that

$$0.75\mathbb{E}[\tilde{x}_2(t)^2] = 1 - 0.1\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)]. \tag{1.4}$$

Moreover, from (1.2) and (1.3), one gets

$$\mathbb{E}[\tilde{x}_2(t+1)\tilde{x}_4(t+1)] = \mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)] = -0.25\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)].$$

Therefore,

$$\lambda_{24} = \mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)] = 0$$

and then from (1.4),

$$\lambda_{22} = \mathbb{E}[\tilde{x}_2(t)^2] = 1/0.75.$$

To obtain λ_{23}, from (1.2) and (1.3), one obtains (note that there is no correlation between $\tilde{v}_1(t)$ and $\tilde{v}_2(t)$)

$$\lambda_{23} = \mathbb{E}[\tilde{x}_2(t+1)\tilde{x}_3(t+1)] = \mathbb{E}[\tilde{x}_2(t)\tilde{x}_3(t)] = -0.5\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)] + 0.1\mathbb{E}[\tilde{x}_4(t)^2] = 0.$$

[8 Marks]

Answers

2. Solution

a) From equations

$$\begin{aligned} x_1(t) &= 0.5x_1(t-1) + x_2(t-1) + v(t), \\ x_2(t) &= -0.1x_2(t-1) + 0.1x_1(t-1), \\ y(t) &= x_1(t), \end{aligned} \tag{2.1}$$

one gets immediately

$$x_1(t+1) = 0.5x_1(t) + x_2(t) + v(t+1)$$

which, since $x_2(t) = -0.1x_2(t-1) + 0.1x_1(t-1)$, can be rewritten as

$$x_1(t+1) = 0.5x_1(t) - 0.1x_2(t-1) + 0.1x_1(t-1) + v(t+1).$$

Moreover, as $x_2(t-1) = x_1(t) - 0.5x_1(t-1) - v(t)$, the ARMA model describing $x_1(t)$ can be rewritten as follows:

$$x_1(t+1) = 0.5x_1(t) - 0.1x_1(t) + 0.05x_1(t-1) + 0.1v(t) + 0.1x_1(t-1) + v(t+1)$$

which by considering $y(t) = x_1(t)$, can be simplified as follows:

$$y(t+1) = 0.4y(t) + 0.15y(t-1) + v(t+1) + 0.1v(t),$$

or

$$y(t) = 0.4y(t-1) + 0.15y(t-2) + v(t) + 0.1v(t-1). \tag{2.2}$$

[3 Marks]

b) First it can be said that the process is stationary as the roots of the characteristic equation of (2.2) in the answer to Question 2-a) $z^2 - 0.4z - 0.15$ are -0.236 and 0.636, and hence they lie inside the unit circle. Moreover, as $\mathbb{E}[v(t)] = 0$, it follows that $\mathbb{E}[y(t)] = 0$. Now, we can obtain $\gamma_y(0)$, $\gamma_y(1)$, and $\gamma_y(2)$ as follows (note that there is no correlation between $y(t)$ and $v(t+\tau), \tau > 0$):

$$\begin{aligned} \mathbb{E}[y(t)^2] =& 0.16\mathbb{E}[y(t-1)^2] + 0.225\mathbb{E}[y(t-2)^2] + 0.01\mathbb{E}[v(t-1)^2] \\ &+ \mathbb{E}[v(t)^2] + 0.12\mathbb{E}[y(t-1)y(t-2)] + 0.08\mathbb{E}[y(t-1)v(t-1)]. \end{aligned}$$

Since the process $y(\cdot)$ is stationary, from (2.2), it follows that

$$\mathbb{E}[y(t)v(t)] = \mathbb{E}[y(t-1)v(t-1)] = 1.$$

Moreover, again by stationarity arguments:

$$\begin{aligned} \gamma_y(0) &= \mathbb{E}[y(t)^2] = \mathbb{E}[y(t-1)^2], \\ \gamma_y(1) &= \mathbb{E}[y(t)y(t-1)] = \mathbb{E}[y(t-1)y(t-2)], \end{aligned}$$

and therefore, it follows that

$$\gamma_y(0) = 0.16\gamma_y(0) + 0.225\gamma_y(0) + 1.01 + 0.12\gamma_y(1) + 0.08$$

which can be simplified as follows:

$$0.615\gamma_y(0) - 0.12\gamma_y(1) = 1.09. \tag{2.3}$$

Answers

Moreover, from (2.2) in the answer to Question 2-a) one gets

$$\mathbb{E}[y(t)y(t-1)] = 0.4\mathbb{E}[y(t-1)^2] + 0.15\mathbb{E}[y(t-1)y(t-2)] + 0.1\mathbb{E}[y(t-1)v(t-1)]$$

which implies that

$$\gamma_y(1) = 0.4\gamma_y(0) + 0.15\gamma_y(1) + 0.1$$

and hence

$$0.4\gamma_y(0) - 0.85\gamma_y(1) = -0.1. \tag{2.4}$$

Therefore, by solution of (2.3) and (2.4), we get

$$\gamma_y(0) = 1.9768, \gamma_y(1) = 1.0479.$$

Finally,

$$\gamma_y(2) = y(t)y(t-2) = 0.4\mathbb{E}[y(t-1)y(t-2)] + 0.15\mathbb{E}[y(t-2)^2].$$

Since the process is stationary,

$$\gamma_y(0) = \mathbb{E}[y(t)^2] = \mathbb{E}[y(t-2)^2],$$

and therefore, it can be said that

$$\gamma_y(2) = 0.4\gamma_y(1) + 0.15\gamma_y(0) = 0.7157.$$

[5 Marks]

c) By considering the ARMA model obtained in the answer to Question 2-a) we write

$$A(z)y(t) = C(z)v(t)$$

where

$$A(z) = 1 - 0.4z^{-1} - 0.15z^{-2},$$
$$C(z) = 1 + 0.1z^{-1}.$$

Therefore,

$$\Gamma_y(\omega) = \frac{|C(z)|^2}{|A(z)|^2}\Big|_{z=e^{j\omega}}.1 = \frac{|z^2+0.1z|^2}{|z^2-0.4z-0.15|^2}\Big|_{z=e^{j\omega}}$$

$$= \frac{(z^2+0.1z)(z^{-2}+0.1z^{-1})}{(z^2-0.4z-0.15)(z^{-2}-0.4z^{-1}-0.15)}\Big|_{z=e^{j\omega}}$$

which can be rewritten as

$$\Gamma_y(\omega) = \frac{1.01+0.1z+0.1z^{-1}}{1.1825-0.34z^{-1}-0.15z^{-2}-0.15z^2-0.34z}$$

$$= \frac{1.01+0.2\cos(\omega)}{1.1825-0.68\cos(\omega)-0.3\cos(2\omega)}.$$

[3 Marks]

Answers

d) By considering $A(z)$ and $C(z)$ obtained in the answer to Question 2-c) we write

$$\hat{y}(t+1|t) = \frac{C(z)-A(z)}{C(z)}y(t+1) = \frac{0.5z^{-1}+0.15z^{-2}}{1+0.1z^{-1}}y(t+1)$$

which can be rewritten as

$$\hat{y}(t+1|t) = \frac{0.5+0.15z^{-1}}{1+0.1z^{-1}}y(t)$$

thus yielding

$$\hat{y}(t+1|t) = -0.1\hat{y}(t|t-1)+0.5y(t)+0.15y(t-1).$$

[3 Marks]

e) By carrying out two steps of the division of $C(z)$ by $A(z)$, the first step gives

$$\frac{C(z)}{A(z)} = \frac{1+0.1z^{-1}}{1-0.4z^{-1}-0.15z^{-2}} = 1+z^{-1}\frac{0.5+0.15z^{-1}}{1-0.4z^{-1}-0.15z^{-2}}$$

and then, in the next step one gets

$$\frac{C(z)}{A(z)} = 1+0.5z^{-1}+z^{-2}\frac{0.35+0.075z^{-1}}{1-0.4z^{-1}-0.15z^{-2}}. \quad (2.5)$$

Therefore, since $v(t) = A(z)/C(z)y(t)$, the optimal two-steps ahead predictor $\hat{y}(t+2|t)$ of $y(t)$ can be written as follows:

$$\hat{y}(t+2|t) = \frac{0.35+0.075z^{-1}}{1+0.1z^{-1}}y(t)$$

which yields

$$\hat{y}(t+2|t) = -0.1\hat{y}(t+1|t)+0.35y(t)+0.075y(t-1).$$

[3 Marks]

f) From (2.5) in the answer to Question 2-e), it follows that

$$\varepsilon(t) = y(t+2)-\hat{y}(t+2|t) = (1+0.5z^{-1})v(t+2)$$

implying

$$\varepsilon(t) = y(t+2)-\hat{y}(t+2|t) = v(t+2)+0.5v(t+1).$$

Thus,

$$\text{var}(\varepsilon) = 1+0.25 = 1.25.$$

[3 Marks]

Answers

3. Solution

a) Based on the discretized form of $\dot{\theta}(t)$ of $\ddot{\theta}(t)$, the discretized form of the model can be immediately obtained as

$$\begin{aligned}\tau(t) =& ml^2(\theta(t+2)-2\theta(t+1)+\theta(t))+c_1(\theta(t+1)-\theta(t))\\ &+c_2\text{sign}(\theta(t))+mlg\cos(\theta(t)).\end{aligned} \tag{3.1}$$

[4 Marks]

b) Since m, l, c_1, c_2 are unknown, the unknown parameters of (3.1) in the answer to Question 3-a) can be chosen as

$$\begin{aligned}\alpha_1 &= ml^2,\\ \alpha_2 &= c_1,\\ \alpha_3 &= c_2,\\ \alpha_4 &= ml.\end{aligned}$$

Therefore, $N = 4$ and accordingly

$$\begin{aligned}\rho_1(t) &= \theta(t+2)-2\theta(t+1)+\theta(t),\\ \rho_2(t) &= \theta(t+1)-\theta(t),\\ \rho_3(t) &= \text{sign}(\theta(t)),\\ \rho_4(t) &= g\cos(\theta(t)).\end{aligned}$$

Thus, the parametric model of $\tau(t)$ can be stated as follows:

$$\tau(t) = \sum_{i=1}^{4}\alpha_i\rho_i(t).$$

[8 Marks]

c) According to the parametric model obtained in the answer to Question 3-b), one considers the estimation of $\tau(t)$ as $\hat{\tau}(t,\hat{\alpha})$ which can be stated as a multiplication of a known nonlinear vector and an unknown parameters vector as follows:

$$\hat{\tau}(t,\hat{\alpha}) = \Psi(t)^\top\hat{\alpha}$$

where the following definitions hold:

$$\Psi(t) = \begin{bmatrix}\theta_m(t+2)-2\theta_m(t+1)+\theta_m(t)\\ \theta_m(t+1)-\theta_m(t)\\ \text{sign}(\theta_m(t))\\ g\cos(\theta_m(t))\end{bmatrix},$$

$$\hat{\alpha} = \begin{bmatrix}\hat{\alpha}_1 & \hat{\alpha}_2 & \hat{\alpha}_3 & \hat{\alpha}_4\end{bmatrix}^\top.$$

To estimate α, we consider the following estimation error:

$$e(j) = \tau_m(j)-\hat{\tau}(t(j),\hat{\alpha}) = \tau_m(j)-\Psi(t(j))^\top\hat{\alpha}, j = 1,2,\ldots,M-2,$$

where $t(j)$ denotes the time associated with the jth sample. Now, we consider a cost function of the form:

$$J(\hat{\alpha}) = \frac{1}{M-2}\sum_{j=1}^{M-2}e(j)^2 = \frac{1}{M-2}\sum_{j=1}^{M-2}\left(\tau_m(j)-\Psi(t(j))^\top\hat{\alpha}\right)^2.$$

Answers

The objective is to obtain the estimated values $\hat{\alpha} = \begin{bmatrix} \hat{\alpha}_1 & \hat{\alpha}_2 & \hat{\alpha}_3 & \hat{\alpha}_4 \end{bmatrix}^\top$ subject to $\min_{\hat{\alpha}} J(\hat{\alpha})$. Therefore, it is sufficient to find $\hat{\alpha}$ making $\frac{\partial J(\hat{\alpha})}{\partial \hat{\alpha}} = [0\ 0\ 0\ 0]$. We have

$$\begin{aligned}\frac{\partial J(\hat{\alpha})}{\partial \hat{\alpha}} &= \frac{2}{M-2}\sum_{j=1}^{M-2}\left(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\right)\frac{\partial}{\partial \hat{\alpha}}\left(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\right) \\ &= -\frac{2}{M-2}\sum_{j=1}^{M-2}\left(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\right)\Psi(t(j))^\top.\end{aligned}$$

Now, to make $\frac{\partial J(\hat{\alpha})}{\partial \hat{\alpha}} = [0\ 0\ 0\ 0]$, we should have

$$\sum_{j=1}^{M-2}\Psi(t(j))\Psi(t(j))^\top \hat{\alpha} = \sum_{j=1}^{M-2}\tau_m(i)\Psi(t(j)).$$

Therefore, if the matrix $\sum_{j=1}^{M-2}\Psi(t(j))\Psi(t(j))^\top$ is full rank, $\hat{\alpha}$ can be obtained as follows:

$$\hat{\alpha} = \left(\sum_{j=1}^{M-2}\Psi(t(j))\Psi(t(j))^\top\right)^{-1}\sum_{j=1}^{M-2}\tau_m(i)\Psi(t(j)).$$

Now, the values of $\hat{l} = \hat{\alpha}_1/\hat{\alpha}_4$, $\hat{c}_1 = \hat{\alpha}_2$, $\hat{c}_2 = \hat{\alpha}_3$, and $\hat{m} = \hat{\alpha}_4/\hat{l}$ can be obtained.

[8 Marks]

Answers

4. Solution

a) A moving average model with order n has a correlation function $\gamma_v(\tau)$ which is zero for $|\tau| > n$. Therefore, we should analyze the empirical estimate of the correlation function as follows:

$$\hat{\gamma}_v(\tau) = \frac{1}{6-\tau}\sum_{i=0}^{5-\tau} v(i)v(i+\tau)$$

yielding

$$\hat{\gamma}_v(0) = 91.065, \hat{\gamma}_v(1) = 13.9, \hat{\gamma}_v(2) = 9.6125, \hat{\gamma}_v(3) = -7.8333, \hat{\gamma}_v(4) = 4.5, \hat{\gamma}_v(5) = 17.$$

Therefore, basing on the available samples, it can be concluded that an MA model cannot model the process.

[4 Marks]

b) The model in prediction form can be stated as

$$\hat{v}(t|t-1) = \hat{a}v(t-1).$$

Now, to estimate a as $\hat{a}$, we consider a cost function for estimation as follows:

$$J(\hat{a}) = \frac{1}{5}\sum_{t=1}^{5}[v(t) - \hat{v}(t|t-1)]^2 = \frac{1}{5}\sum_{t=1}^{5}[v(t) - \hat{a}v(t-1)]^2.$$

Therefore, the optimal value $\hat{a}$ is the argument of the minimum of the cost function which can be obtained as follows:

$$\frac{\partial J(\hat{a})}{\partial \hat{a}} = -\frac{2}{5}\sum_{t=1}^{5}[v(t)v(t-1) - \hat{a}v(t-1)^2] = 0$$

thus getting

$$\hat{a} = \frac{\sum_{t=1}^{5} v(t)v(t-1)}{\sum_{t=1}^{5} v(t-1)^2} = \frac{69.5}{543.5} = 0.1279.$$

[6 Marks]

c) The model in the prediction form can be stated as

$$\hat{v}(t|t-1) = \hat{a}v(t-1) + \hat{b}v(t-2).$$

Now, to estimate a as $\hat{a}$ and b as $\hat{b}$, we consider a cost function for estimation as follows:

$$J(\hat{a},\hat{b}) = \frac{1}{5}\sum_{t=2}^{5}[v(t) - \hat{v}(t|t-1)]^2 = \frac{1}{5}\sum_{t=2}^{5}[v(t) - \hat{a}v(t-1) - \hat{b}v(t-2)]^2.$$

Therefore, the optimal $(\hat{a},\hat{b})$ is the argument of the minimum of the cost function which can be obtained as follows:

$$\frac{\partial J(\hat{a},\hat{b})}{\partial \hat{a}} = -\frac{2}{5}\sum_{t=2}^{5}[v(t)v(t-1) - \hat{a}v(t-1)^2 - \hat{b}v(t-1)v(t-2)] = 0,$$

$$\frac{\partial J(\hat{a},\hat{b})}{\partial \hat{b}} = -\frac{2}{5}\sum_{t=2}^{5}[v(t)v(t-2) - \hat{a}v(t-1)v(t-2) - \hat{b}v(t-2)^2] = 0.$$

Answers

Hence:

$$-130.5 - 443.5\hat{a} - 73.75\hat{b} = 0,$$
$$38.45 - 73.75\hat{a} - 537.25\hat{b} = 0.$$

Therefore, one gets

$$\hat{a} = -0.3133, \hat{b} = 0.1146.$$

[6 Marks]

d) Performance indexes can be taken as the cost functions derived in the answers to Questions 4-b) and 4-c) evaluated on the calculated optimal values of the parameters that is: $J(\hat{a})$ for $\hat{a} = 0.1279$ and $J(\hat{a}, \hat{b})$ for $(\hat{a}, \hat{b}) = (-0.3133, 0.1146)$. Hence,

$$J(0.1279) = 87.5,$$
$$J(-0.3133, 0.1146) = 0.2746.$$

These values of the performance indexes clearly show the improvement using a higher-order model.

[4 Marks]

Answers